

Code No: 155DB**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, January/February - 2023****SOFTWARE ENGINEERING****(Common to CSE, IT, ECM, ITE)****Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) What are software engineering layers? [2]
- b) Write about process assessment. [3]
- c) Define procedural interfaces. [2]
- d) Discuss about requirements validation in brief. [3]
- e) What is UML? Write the principles of modelling. [2]
- f) Give an example of a class diagram. [3]
- g) Define white-box testing and black –box testing. [2]
- h) How debugging differs from testing? [3]
- i) What is software risk? [2]
- j) What are the metrics for software quality? [3]

PART – B**(50 Marks)**

- 2.a) Define software. Explain in detail about software myths.
 - b) Discuss in detail about water fall process model. [5+5]
- OR**
- 3.a) What is a process model? Explain about prototyping model in detail.
 - b) What is CMMI? Explain about CMMI levels. [5+5]
- 4.a) Explain about requirements management phases of requirement engineering process.
 - b) Explain about state machine models with examples. [5+5]
- OR**
- 5.a) Based on your experience with a bank ATM, draw a data-flow diagram modelling the data processing involved when a customer withdraws cash from the machine.
 - b) Discuss about architectural design in brief. [5+5]
- 6.a) What are building blocks of the UML? Explain.
 - b) Explain about refining the architecture into components. [5+5]
- OR**
- 7.a) Distinguish between sequence and collaboration diagrams.
 - b) Briefly explain about the design model. [5+5]

- 8.a) What is integration testing? Explain in detail.
b) Discuss about metrics for testing in detail. [5+5]

OR

- 9.a) Distinguish between verification and validation. Explain about organizing for software testing.
b) Explain about the metrics for design model. [5+5]

10. Explain the following:
a) Software quality concepts
b) Risk identification. [5+5]

OR

11. Explain the following:
a) Statistical SQA
b) Developing a risk table. [5+5]

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Code No: 155DB**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, July/August - 2023****SOFTWARE ENGINEERING****(Computer Science and Engineering - IOT)****Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) What is product-line software? [2]
- b) Write down the pros and cons of the concurrent development model. [3]
- c) What are the functional requirements and non-functional requirements? [2]
- d) Discuss about feasibility report in brief. [3]
- e) How to represent class, interface, object and package in UML? [2]
- f) What are issues and challenges in the design process? [3]
- g) What is Unit testing? [2]
- h) What are the attributes of effective software metrics? [3]
- i) Define FP metric. [2]
- j) Write about SQA activities. [3]

PART – B**(50 Marks)**

- 2.a) Discuss about the changing nature of the software.
- b) Explain about the unified process in detail. [5+5]

OR

- 3.a) What is RAD model? Explain with RAD model with neat diagram.
- b) Distinguish between TSP and PSP models. [6+4]

- 4.a) Discuss the problems of using natural language for defining user and system requirements, and show, using small examples, how structuring natural language into forms can help avoid some of these difficulties.
- b) Explain in detail about requirements elicitation and analysis process. [5+5]

OR

- 5.a) Using your knowledge of how an ATM is used, develop a set of use-cases that could serve as a basis for understanding the requirements for an ATM system.
- b) Describe the structure of a software requirements document. [5+5]

- 6.a) Explain terms and concepts of sequence diagrams.
b) Give a brief note on taxonomy of architectural styles. [5+5]

OR

- 7.a) What are design concepts? Explain in detail.
b) Draw use case diagrams of ATM system and explain. [5+5]

- 8.a) What is debugging? Explain about the debugging process.
b) Explain various metrics for testing and debugging process. [5+5]

OR

- 9.a) Discuss in detail about validation testing.
b) Explain about component-level design metrics. [5+5]

10. Explain the following:
a) Risk projection
b) The ISO 9000 quality standards. [5+5]

OR

11. Explain the following:
a) RMMM plan
b) Software Quality Matrix. [5+5]

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Code No: 155DB**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, August/September - 2024****SOFTWARE ENGINEERING****(Computer Science and Engineering - IOT)****Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) What is the significance of software engineering in today's technology-driven world? [2]
- b) How does the concept of continuous improvement apply to software engineering processes? [3]
- c) How do object models differ from other types of system models? [2]
- d) How do functional requirements contribute to defining the behavior of a software system? [3]
- e) What is a software architecture, and why is it important in software development? [2]
- f) Identify two key components depicted in a collaboration diagram and explain their roles. [3]
- g) What is cyclomatic complexity, and how does it relate to software maintainability? [2]
- h) What is the difference between top-down and bottom-up integration testing approaches? [3]
- i) Describe the difference between direct and indirect measures in software metrics. [2]
- j) Differentiate between reactive and proactive risk management strategies in software projects. [3]

PART – B**(50 Marks)**

- 2.a) Describe the Capability Maturity Model Integration (CMMI) and its significance in software engineering.
 - b) Explain the unified process and its key characteristics. [5+5]
- OR**
- 3.a) Explain the waterfall model of software development, highlighting its advantages and limitations.
 - b) Describe the role of feedback loops in improving software development processes. [5+5]

- 4.a) Explain the difference between user requirements and system requirements.
b) Write a software requirement specification document for an online customer service web portal for an online cosmetic company. [5+5]

OR

- 5.a) Describe techniques for prioritizing and managing conflicting requirements during the requirements engineering process.
b) Describe various techniques for requirements elicitation and analysis. [5+5]
- 6.a) Discuss the importance of design quality in software development.
b) How do sequence diagrams illustrate the flow of messages and interactions between objects over time? Give an example. [5+5]

OR

- 7.a) How do component diagrams represent the physical components and dependencies of a software system? Discuss their role.
b) Create a class diagram for a simple library management system, including classes for books, library members, and library staff. [5+5]
- 8.a) Discuss various validation techniques and their relevance in different contexts.
b) What are the metrics commonly used for evaluating the effectiveness of testing processes? [6+4]

OR

- 9.a) Compare and contrast unit and integration testing techniques.
b) Describe metrics commonly used for assessing the design model in software development. [4+6]
- 10.a) Evaluate the significance of software quality metrics in assessing the overall quality of software products.
b) Explain the concept of risk identification in software projects. [5+5]

OR

- 11.a) Explain the ISO 9000 quality standards and their relevance to software engineering.
b) Describe techniques for measuring and improving software reliability. [5+5]

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JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech III Year I Semester Examinations, March - 2024

SOFTWARE ENGINEERING

(Common to CSE, IT, ECM, ITE, AI&DS, CSD)

Time: 3 Hours

Max. Marks: 75

Note: i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

- 1.a) What do you mean by software myths? [2]
- b) What is process assessment, and how is it conducted in software engineering? [3]
- c) What are structured methods in the context of system modelling? [2]
- d) How do requirements management tools and techniques facilitate the organization and tracking of requirements throughout the software development lifecycle? [3]
- e) How does UML facilitate the representation and visualization of software architectures? [2]
- f) How do class diagrams depict relationships and interactions among classes in a system? [3]
- g) What do you understand by equivalence partitioning in software testing? [2]
- h) Differentiate between validation testing and verification testing in software engineering. [3]
- i) Define software reliability. [2]
- j) Define software quality assurance and its role in software development processes. [3]

PART – B

(50 Marks)

- 2.a) Discuss incremental process models and provide examples of their application in real-world projects.
- b) Compare and contrast personal and team process models in software development. [5+5]

OR

- 3.a) What are evolutionary process models, and how do they differ from traditional approaches like the waterfall model?
- b) Explain how process models facilitate collaboration and communication among project stakeholders. [5+5]
- 4.a) Define functional and non-functional requirements in software engineering. Provide examples of each.
- b) Write a software requirement specification document for the Android to-do list application. [5+5]

OR

- 5.a) What role does validation play in ensuring the correctness and completeness of software requirements? Discuss different validation techniques.
- b) What is a feasibility study, and why is it conducted during the requirements engineering process? Discuss the key factors considered in a feasibility study. [5+5]

- 6.a) Explain the design process in software engineering. What are the key phases involved, and how do they contribute to the overall development lifecycle?
- b) Discuss the role of software architects in defining and implementing architectural designs. [5+5]

OR

- 7.a) Describe the design model in software engineering. How does it serve as a blueprint for implementing and maintaining software systems?
- b) Describe basic structural modelling techniques in UML. [5+5]

- 8.a) Compare and contrast black-box and white-box testing techniques. Provide examples of scenarios where each technique would be most appropriate.
- b) Evaluate the effectiveness of using cyclomatic complexity as a metric for measuring the complexity of source code. [5+5]

OR

- 9.a) Describe common debugging techniques and strategies for identifying and fixing software defects efficiently.
- b) Define software quality metrics and explain why they are essential for assessing the overall quality of software products. [5+5]

- 10.a) Define software measurement and explain its importance in the software development process.
- b) Discuss the differences between reactive and proactive risk management strategies. Provide examples of situations where each strategy would be most appropriate. [5+5]

OR

- 11.a) Explain the concept of Risk Mitigation, Monitoring, and Management (RMMM) in software engineering.
- b) Discuss the benefits of conducting FTRs and the roles of stakeholders involved. [5+5]

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